What representations guide learning to process singular *they* efficiently?

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Recently the English pronoun system has re-organized to integrate a new use of singular *they* that refers to a person whose personal pronouns are they/them, e.g. “Alex went out. They saw a movie”. We call this form “personal *they*” (elsewhere called “nonbinary *they”*). People vary widely in whether they rate personal *they* as acceptable (Conrod et al., 2022) and it can induce processing difficulty (Leventhal et al., 2020). Yet remarkably, people achieve relative success in producing personal *they* (Arnold et al., 2024) and can assign a singular interpretation in supporting contexts (Arnold et al., 2021). This is especially notable because at this point in time, nearly all adults can remember a time when they had never heard of the form. By what mechanism do adults gain fluency with this new usage?

Research reveals two possible mechanisms. First, increased usage of singular *they* is considered a “change from above” (Conrod, 2018), supported by social awareness of inclusivity. But social intentions alone do not create automatic fluency with the form. Second, recent findings show that people are more likely to take a singular interpretation, and they do so more quickly, if they have recently been exposed to examples of personal *they* (Ye & Arnold, 2025). This is an example of adaptation, where linguistic structures become more available through exposure (e.g., Bock, 1986; Fine & Jaeger, 2013).

But an open question is how adaptation occurs. Imagine that I hear several sentences using personal *they* to refer to Alex, who uses they/them pronouns. What linguistic representations does this activate? Possibility 1 is that it activates the specific link between *they* and Alex, for example serving as a reminder that Alex uses they/them pronouns. If so, adaptation should only make it easier to understand future examples of *they* referring to Alex. Possibility 2 is that it additionally activates the abstract representation that *they* can be used in the singular for reference to a specific individual who uses they/them pronouns. If so, exposure to examples of personal *they* referring to Alex should generalize to personal *they* for other people.

We use the same experimental paradigm as Ye & Arnold (2025), who demonstrated that adaptation facilitates the interpretation of personal *they.* In their study, participants met three characters and learned their pronouns: Alex (*they*), Liz (*she*) and Will (*he*). Then they heard 24 exposure stories with either singular or plural *they* (Fig. 1). Their task was to click on the object mentioned (sandwich/sandwiches). Participants viewed a display with pictures of Liz, Alex, and Liz-and-Alex (Fig. 1). They were told that each character interacts with the objects under their picture. Half the participants saw only singular exposure stories, where the object always appeared under Alex; half saw only plural exposure stories, where the object always appeared under the joint picture. The critical question was how they responded to the 20 interspersed ambiguous stories where there were two matching object pictures (Fig. 2). Which object they clicked on reflected their interpretation of *they.* The singular selection was more common in the singular exposure condition (53%) than plural (27%). Reaction time revealed an interaction between priming and selection: singular responses were faster in the singular exposure condition, and plural in the plural exposure condition.

Our current question is whether exposure to *they* for Alex (the primed character) also affects interpretation of *they* referring to two new (unprimed) characters, Wren and Dana, who use they/them pronouns. The 24 exposure stories (6 appearing before any critical stories) all referred to Alex in the singular condition. The critical stories included 10 Alex (primed character) stories, and 10 Dana/Wren (unprimed character) stories. (Fig. 3). **Results:** People chose the singular more in the singular than plural priming conditions for both primed and unprimed characters (Fig. 4). Reaction times were faster for plural than singular in the plural priming condition, but singular priming eliminated this difference (Fig. 5). **Conclusions**: Findings support the role of adaptation to growing facility with understanding personal *they* and show that exposure is represented broadly and generalizes to all referents who use *they/them* pronouns.

Fig. 1. Example exposure story “Liz and Alex ordered lunch at the cafeteria. Liz showed the menu to Alex. Then they asked for a sandwich” (left, singular) or “…sandwiches (right, plural).

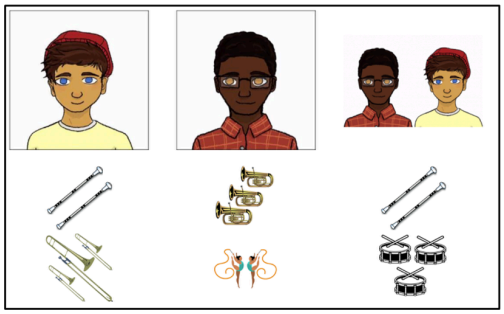

Fig. 2. Example critical Alex (primed) story (“Will and Alex were at marching band practice. Will gave some water to Alex. Then they put down the batons”).

Fig. 3. Example critical Dana (unprimed) story (“Will and Dana went to play soccer. Will handed some cones to Dana. Then they put on their cleats.”

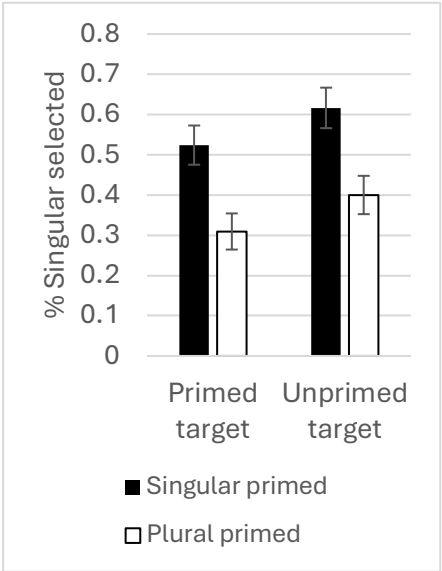

Fig. 4. Rate of selecting the singular interpretation was greater in the singular vs. plural prime condition (b = 1.5 (0.5), z = 3.1, p = .002)

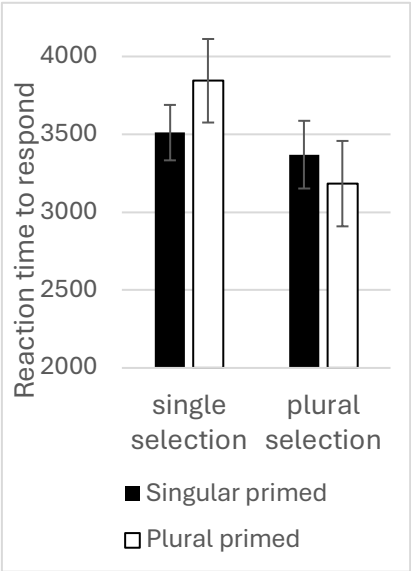

Fig. 5. Reaction time results: priming and selection interact (b = -0.07 (0.03), t = -2.3, p= .03). (Shown here as raw RT; analyzed as logRT)

References: Arnold et al. (2021). PBR. ♦ Arnold et al. (2024). *Glossa Psycholx*. ♦ Ye & Arnold. (2025). HSP. ♦ Bjorkman, (2017). *Glossa.* ♦ Bock (1986). *Cog. Psych.* ♦ Bradley et al. (2019). Personality…ET. ♦ Conrod (2018). Changes in singular they. ♦ Conrod, K. (2020). The Oxford Handbook ... ♦ Conrod, K., et al. (2022). NELS 52. ♦ Fine & Jaeger (2013). *Cognitive Science.* ♦ Leventhal et al., (2020). CUNY 2020.